

Topic 1 — Independent Practice Part 1 Solutions

Brief checks. Use **We Need** → **We Know** → **We Solve** in your notebook.

Topic 1-1 Relate Integers and Opposites

Question 1

We Need: The additive inverse of a 4.5 m climb.

We Know: Additive inverse means same distance, opposite direction.

We Solve: Go 4.5 meters down (toward / below the starting level).

Answer: 4.5 meters downward

Question 2

We Need: An equation with $|a|$ and $|b|$.

We Know: $a + b = 0$ means $b = -a$, so the distances from zero match.

We Solve: $|a| = |b|$

Answer: $|a| = |b|$ (or equivalent)

Question 3

We Need: $|-9|$

We Know: Absolute value is distance from zero.

We Solve: $|-9| = 9$

Answer: 9

Question 4

We Need: Order -3 , 8 , -1 least to greatest.

We Know: On a number line, more left means smaller.

We Solve: -3 , -1 , 8

Answer: -3 , -1 , 8

Topic 1-2 Understand Rational Numbers

Question 1

We Need: Decimal for $\frac{3}{8}$.

We Know: Divide $3 \div 8$.

We Solve: $3 \div 8 = 0.375$

Answer: 0.375

Question 2

We Need: Which statement is *not* right.

We Know: A fraction (with nonzero denominator) becomes a decimal that terminates or repeats.

We Solve: C is not right.

Answer: C

Question 3

We Need: Who is correct about $0.\overline{27}$, Jordan or Casey.

We Know: Let $x = 0.\overline{27}$. Then $100x - x = 99x$, and the repeating tails cancel.

We Solve: $100x = 27.\overline{27}$, so $99x = 27$ and $x = \frac{27}{99} = \frac{3}{11}$. A repeating decimal is rational.

Answer: Jordan; $0.\overline{27} = \frac{3}{11}$

Topic SC-1 Compare Rational Numbers

Question 1

We Need: Two comparisons of 10.4 and 10.6.

We Know: On a number line, more right means greater.

We Solve: $10.4 < 10.6$ and $10.6 > 10.4$

Answer: $10.4 < 10.6$ and $10.6 > 10.4$ (pounds OK)

Question 2

We Need: Symbols that fit $\frac{3}{4}$ and 0.75.

We Know: $\frac{3}{4} = 0.75$

We Solve: Equal numbers also satisfy $\leq$ and $\geq$.

Answer: $=, \leq, \geq$

Topic 1-3 Add Integers

Question 1

Part A: Start at 0. Move right 10, left 4, left 8.

Part B: $10 + (-4) + (-8) = -2$

Answer: Part A: number-line moves as above; Part B: -2

Question 2

We Need: The sum of the three signed moves.

We Know: Combine the positives and negatives.

We Solve: $15 + (-20) + 3 = -2$

Answer: -2

Question 3

We Need: Balance after 6 months.

We Know: Fee \$12.50 each month; $6 \times \$12.50 = \75 .

We Solve: $\$500 - \$75 = \$425$

Answer: \$425

Question 4

We Need: The sum.

We Know: Opposite signs: subtract the distances, keep the sign of the larger.

We Solve: $-12 + 19 = 7$

Answer: 7

Topic 1-4 Subtract Integers

Question 1

We Need: Distance from 5°F to -8°F .

We Know: Distance $= |5 - (-8)| = 13$.

We Solve: Valid expressions include $|5 - (-8)|$, $|-8 - 5|$, $5 - (-8)$, $|-8| + 5$.

Answer: Distance 13°F ; any correct distance expressions

Question 2

We Need: Distance between the two heights.

We Know: Distance is the absolute value of the difference.

We Solve: $|150 - 120| = 30$. Valid: $|150 - 120|$, $|120 - 150|$, $150 - 120$.

Answer: Distance 30 ft; any correct distance expressions

Question 3

We Need: Distance between the two depths.

We Know: Distance is the absolute value of the difference.

We Solve: $|-25 - (-40)| = 15$. Valid: $|-25 - (-40)|$, $|-40 - (-25)|$, $40 - 25$.

Answer: Distance 15 ft; any correct distance expressions

Question 4

We Need: Total distance the elevator travels.

We Know: Distance between two floors is the absolute value of their difference; add the legs.

We Solve: $|8 - (-3)| = 11$ and $|-1 - 8| = 9$; $11 + 9 = 20$.

Answer: 20 floors

Topic 1-5 Add and Subtract Rational Numbers

Question 1

We Need: Balance after 4 months.

We Know: Start \$200 (below \$250), so the fee applies each month.

We Solve: $200 - 15 = 185$; $185 - 15 = 170$; $170 - 15 = 155$; $155 - 15 = 140$.

Answer: \$140

Question 2

We Need: Temperature after the three drops.

We Know: Each drop is a negative change; add them to the start.

We Solve: $-3.75 - 2.4 - 1.85 = -8.00$

Answer: -8°C

Question 3

We Need: Distance between the two marks.

We Know: Distance is the absolute value of the difference; write the mixed number as a decimal first.

We Solve: $2\frac{1}{2} = 2.5$; $|2.5 - (-1.75)| = 4.25$.

Answer: 4.25 miles

Topic 1-6 Multiply Integers

Question 1

Part A: Descent is negative: -15 feet per minute.

Part B: $(-15) \times 12 = -180$ feet (180 feet below sea level).

Answer: Part A: -15 ; Part B: -180 feet (below sea level)

Question 2

We Need: Depth after 4.5 minutes.

We Know: A descent rate is negative, so the product is negative.

We Solve: $(-250.50) \times 4.5 = -1127.25 \text{ m} = -1.12725 \text{ km} \approx -1 \text{ km}$.

Answer: -1 kilometer (nearest km; descent)

Question 3

We Need: $(-5)^2$

We Know: The parentheses mean the negative is part of the base.

We Solve: $(-5)^2 = (-5) \times (-5) = 25$

Answer: 25

Question 4

We Need: -5^2

We Know: Without parentheses, the exponent applies to 5 only.

We Solve: $-5^2 = -(5^2) = -25$

Answer: -25

Difference: parentheses square the negative; without them, square first, then make negative.

Topic 1-7 Multiply Rational Numbers

Question 1

We Need: The product.

We Know: Do the parentheses first, then multiply.

We Solve: $-0.5(10 + 6) = -0.5 \times 16 = -8$

Answer: -8

Topic 1-8 Divide Integers

Question 1

Part A: Unit rate $= (-168) \div 7 = -24$ feet per minute.

Part B: $(-24) \times 4 = -96$ feet (96 feet below sea level).

Answer: Part A: -24 ; Part B: -96 feet (below sea level)

Question 2

We Need: The quotient.

We Know: Opposite signs $\Rightarrow$ negative quotient.

We Solve: $(-96) \div 8 = -12$

Answer: -12

Question 3

We Need: The quotient.

We Know: Same signs $\Rightarrow$ positive quotient.

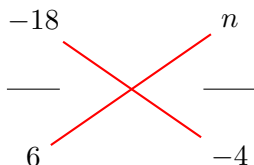
We Solve: $(-72) \div (-9) = 8$

Answer: 8

Question 4

We Need: n in $\frac{-18}{6} = \frac{n}{-4}$.

We Know: The fractions are equal. Use the butterfly method.



Butterfly: multiply along each red line.

We Solve: Cross products: $(-18)(-4) = 6 \cdot n$, so $72 = 6n$ and $n = 12$.

Check: $(-18) \div 6 = -3$ and $12 \div (-4) = -3$.

Answer: $n = 12$

Topic 1-9 Divide Rational Numbers

Question 1

We Need: Each owner's share of the loss.

We Know: Split the loss equally among the 3 owners.

We Solve: $(-450.75) \div 3 = -150.25$

Answer: -150.25 (or $-\$150.25$ per owner)

Question 2

We Need: How many water stations.

We Know: $15\frac{1}{2} = \frac{31}{2}$, $3\frac{7}{8} = \frac{31}{8}$.

We Solve: $\frac{31}{2} \div \frac{31}{8} = \frac{31}{2} \times \frac{8}{31} = 4$ intervals. Stations at the end of each interval (including the finish line) $\Rightarrow 4$.

Answer: 4 water stations

Topic 1-10 Solve Problems with Rational Numbers

Question 1

We Need: The difference in elevation.

We Know: Subtracting a negative adds its opposite.

We Solve: $450 - (-120) = 450 + 120 = 570$

Answer: 570 feet

Topic SC-2 Use the Laws of Exponents

Question 1

We Solve: $3^2 \times 3^4 = 3^{2+4} = 3^6 = 729$

Answer: 729 square inches

Question 2

Part A: $\frac{2^4}{2^2}$ or 2^{4-2} or 2^2

Part B: $16 \div 4 = 4$

Answer: Part A: $\frac{2^4}{2^2}$ (or equivalent); Part B: 4 cupcakes per friend

Question 3

We Solve: $(4^3)^2 = 4^{3 \cdot 2} = 4^6 = 4096$

Answer: 4096

Question 4

We Solve: $7^0 = 1$

Answer: 1

Question 5

We Solve: $\frac{4^5}{4^2} \cdot 4^0 \cdot 4^{-1} = 4^{5-2} \cdot 1 \cdot \frac{1}{4} = 4^3 \cdot \frac{1}{4} = 64 \div 4 = 16$

Answer: 16

Question 6

We Solve: $(2^3)^2 = 2^6 = 64$, $(3 \cdot 2)^2 = 6^2 = 36$, and $\left(\frac{12}{2}\right)^2 = 6^2 = 36$. Then $64 \cdot 36 \div 36 = 64$.

Answer: 64

Question 7

We Solve: $5^0 = 1$, $3^{-2} = \frac{1}{9}$, and $\frac{2^6 \cdot 2^3}{2^7} = 2^{6+3-7} = 2^2 = 4$. Thus $1 + \frac{1}{9} + 4 = 5\frac{1}{9}$.

Answer: $\frac{46}{9}$ or $5\frac{1}{9}$

Question 8

We Solve: $\frac{7^4}{7^2} \cdot 7^0 = 7^{4-2} \cdot 1 = 7^2 = 49$

Answer: 49

Question 9

We Solve: $(3^2)^3 = 3^{2 \cdot 3} = 3^6 = 729$

Answer: 729

Question 10

We Solve: $(2 \cdot 5)^3 = 10^3 = 1000$

Answer: 1000

Question 11

We Solve: $6^{-2} = \frac{1}{36}$ and $\left(\frac{8}{4}\right)^3 = 2^3 = 8$. Then $\frac{1}{36} + 8 = \frac{1}{36} + \frac{288}{36} = \frac{289}{36} = 8\frac{1}{36}$.

Answer: $\frac{289}{36}$ or $8\frac{1}{36}$

Question 12

We Solve: $3^4 \cdot 3^2 = 3^{4+2} = 3^6 = 729$

Answer: 729